Arkaprava Bose

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Education

Department of Physical Sciences, IISER Kolkata — West Bengal, India
5-Year BSMS Dual Degree, Minor in Statistics (in progress) — 2023 - 2028 *(expected)*

- CGPA: 9.05/10
- Minor coursework: Statistical Inference, Stochatic Processes, Probability & Statistics I & II, Linear Algebra I, Analysis I & II

Cambridge School, Rewari — Haryana, India
Higher Secondary & Secondary Education — 2013 - 2023

- Higher Secondary Certification (2023): 95.2%
- Secondary Certification (2021): 94%

Research Projects

Stability of Modular Networks | Kolkata, India — Aug 2026 | *Ongoing*
Guide: Dr. Anandamohan Ghosh
Professor,Department of Physical Sciences, IISER Kolkata

- Spectral analysis of modular random community matrices to predict ecological stability

Response Shifts and Tipping in Nonlinear Maps | Remote — June 2026 | *Ongoing*
Guide: Prof. Marc Timme
Professor, TU Dresden

- Analyzed average response shifts and tipping in periodically driven nonlinear maps using perturbation theory, bifurcation analysis and numerics.

Synchronization in Chaotic dynamics on network | Chennai, India — May 2026 - June 2026 | *Completed*
Guide: Prof. Sitabhra Sinha
Professor, The Institute of Mathematical Sciences (IMSc)

- Numerical study of chaotic synchronization in 1-D and 2-D lattices with nearest neighbor coupling, with simulations in MATLAB.

Relaxation Rates in Multi-Qubit Systems | Mumbai, India — Dec 2024 - May 2026 | *Completed*
Guide: Dr. Sudhir R Jain
Professor of Eminence, Somaiya Vidyavihar University

- Studied relaxation and excitation of two interacting superconducting qubits coupled to a driven nonlinear Duffing oscillator.

NIUS 21.1 | Mumbai, India — June 2024 | *Completed*

- Physics programme on superconductors, quantum mechanics, neural networks and quantum computing, with an experimental problem-solving lab course and visits to TIFR research labs.

Scholarships & Awards

- **INSPIRE Scholarship** — 2023 - 2028
  Department of Science and Technology, Govt. of India

Skills

Programming: `Python` (proficient), `MATLAB` (familiar), `C` (familiar)

Scientific Computing: `NumPy`, `SciPy`, `Matplotlib`; numerical continuation, spectral computations, direct simulation

Research Methods: nonlinear dynamics and bifurcation theory, random matrix theory, perturbation theory, stochastic processes, statistical mechanics

Tools: Git/GitHub, LaTeX, Jupyter

Languages: English, Hindi, Bengali

Interests: Badminton

Outreach & Leadership

- **Outreach Volunteer and Educator**, Ek Pehal (IISER Kolkata) DATES
  Taught underprivileged children in nearby villages and on campus through hands-on, activity-based sessions.
- **Event Organizer**, Inquivesta (IISER Kolkata's annual fest) DATES
  Organized LOST, the fest's flagship treasure hunt: designed puzzles and game mechanics, and managed budget, publicity and participants.